

Chemical Bond

What is chemical bond?

A chemical bond is a strong attractive force that exists between certain atoms in a substance.

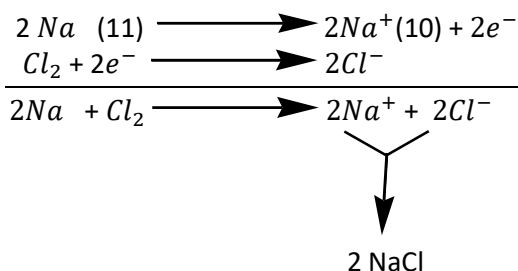
Types of chemical bond

- a) Ionic bond
- b) Covalent bond
- c) Co-ordinate covalent bond

a) Ionic bond:

An ionic bond is a chemical bond formed by the electrostatic attraction between positive and negative ions.

The bond forms between two atoms when one or more electrons are transferred from the valence shell of one atom to the valence shell of the other. The atom that loses electrons becomes a **cation** (positive ion) and the atom that gains electrons becomes an **anion** (negative ion). Example:



Another example: KCl, CaCl₂, Na₂SO₄, CuSO₄, AlCl₃, K₂SO₄ etc.

In ionic bond metal and nonmetal must be present.

b) Covalent bond:

A covalent bond is a chemical bond formed by the sharing of a pair of electrons between atoms. Example: CO₂, H₂O, HCl, HNO₃, H₂SO₄ etc. There are two types of covalent bonds:

In covalent bond nonmetal must be present.

i) **Non polar covalent bond:**

A covalent bond between two similar atoms which have the same electronegativity or zero electronegativity difference is called a non-polar or homo polar covalent bond. e.g. H₂, O₂, N₂, Cl₂ etc.

ii) **Polar covalent bond:**

A covalent bond between two dissimilar atoms which have different electronegativity values is called a polar covalent bond. e.g. H₂O, HCl, HF etc.

Percentage of ionic character in a polar covalent bond:

Relation between $(x_A - x_B)$ values, percentage ionic character in A-B bond and the nature of A-B bond

$(x_A - x_B)$	% ionic character in A-B bond	Nature of A-B bond and its representation	Examples of bonds
0	0	Purely covalent(A-B)	$N \equiv N, O = O, Cl - Cl, H - H$
0.1-0.8	0.5-15	Covalent(A-B)	$C - S, C - I, N - Cl, P - H$
0.9-1.6	17-19	Polar covalent($A^{\delta-} - B^{\delta+}$)	Both $O - H$ bond in H_2O molecule are polar covalent bond and are represented as $O^{\delta-} - H^{\delta+}$
1.9	50	50% ionic & 50% covalent	-
1.9-2.0	55-63	Ionic ($A^- - B^+$)	-
2.0-2.3	67-74	Ionic ($A^- - B^+$)	$Na - Cl (Na^+ - Cl^-)$
2.3-3.3	70-93	Ionic ($A^- - B^+$)	$Cs - F (Cs^+ - F^-)$

The following points may be noted from the above table:

- i) When $(x_A - x_B) = 1.9$, the amount of ionic character in A-B bond is 50%. Thus A-B bond is 50% ionic and 50% covalent.
- ii) When $(x_A - x_B) < 1.9$, the amount of ionic character in A-B bond is less than 50% & that of covalent character is more than 50%. Thus A-B bond is predominantly covalent and hence is represented as A- B.
- iii) When $(x_A - x_B) > 1.9$, the amount of ionic character in A-B bond is more than 50% & that of covalent character is less than 50%. Thus A-B bond is predominantly ionic and hence is represented as $A^- - B^+$.

Henny and Smyth Equation: % ionic character in $A - B$ bond = $[16(x_A - x_B) + 3.5 (x_A - x_B)^2] \%$

Q-1: Calculate the percentage of ionic character in $Cs - F$ bond in CsF molecule. The Electronegativity of Cs and F are 0.7 and 4.0 respectively. Predict the nature of CsF molecule.

Answer:

$$(x_A - x_B) = x_F - x_{Cs} = 4.0 - 0.7 = 3.3$$

$$\begin{aligned} \text{\% ionic character in } Cs - F \text{ bond} &= [16 \times 3.3 + 3.5 \times (3.3)^2] \% \\ &= 90.9\% \end{aligned}$$

Since the ionic character is more than 50 %, CsF molecule is a purely ionic molecule.

Q-2: Calculate the percentage of ionic character in $K - Cl$ bond in KCl molecule. The Electronegativity of K and Cl are 0.8 and 3.0 respectively. Predict the nature of KCl molecule.

Q-3: Arrange the molecules HF, HCl, HBr & HI in the decreasing order of the percentage of ionic character in $H - X$ bond ($X = F, Cl, Br$ or I), Electronegativity values are as: $F = 4.0, Cl = 3.0, Br = 2.8, I = 2.2$ and $H = 2$.

% ionic character in H- F bond in HF molecule =

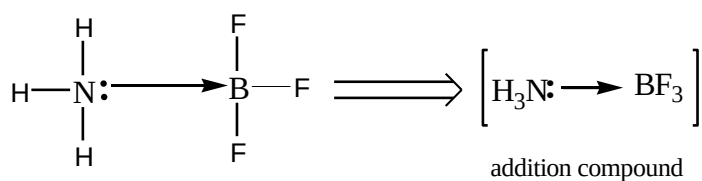
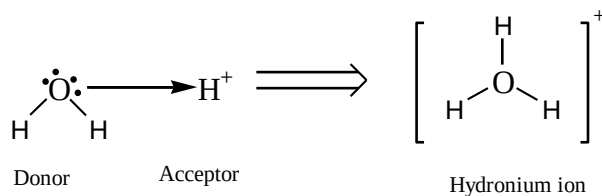
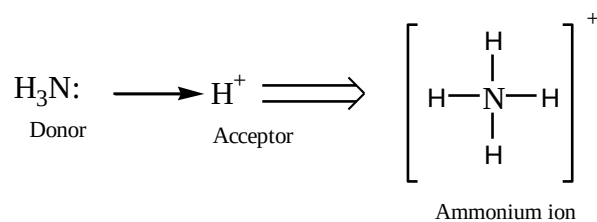
% ionic character in H- Cl bond in HCl molecule =

% ionic character in H- Br bond in HBr molecule =

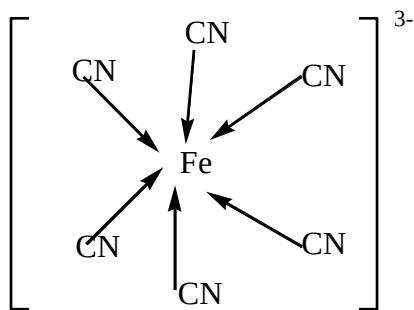
% ionic character in H- I bond in HI molecule =

c) Co-ordinate covalent bond:

A co-ordinate covalent bond is a bond formed when both electrons of the bond are donated by one atom.

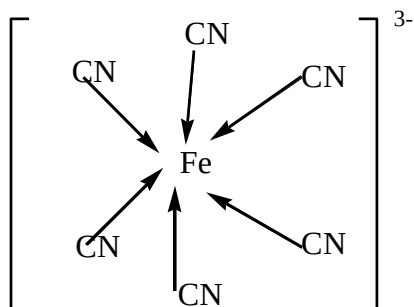


Co-ordination compound: Co-ordination compounds in which a particularly charged central metal ion (or a neutral metal atom) attached to a surrounding group (ions or molecules) by coordinate bonds.
e.g.



Ligand: Ions or molecules which attached to a central atom by co-ordinate bonds are called ligand. e.g.
 CN^- , Cl^- , Br^- , H_2O , NH_3 etc.

Co-ordination no.: The co-ordination no. of a metal atom in complex is the total no. of bonds the metal atom form with ligands. e.g.



The co-ordination no. of this complex is 6.

Hybridization:

Hybridization is the combination of atomic orbitals like s,p,d or f to give the equivalent orbitals which are known as hybrid orbitals. This phenomenon is known as Hybridization.

Hybridization is actually a process in which atomic orbitals having different shape and energy intermix to form a set of same number of orbitals which have same shape and energy.

Question: What is hybridization? Mention two characteristics of hybrid orbital.

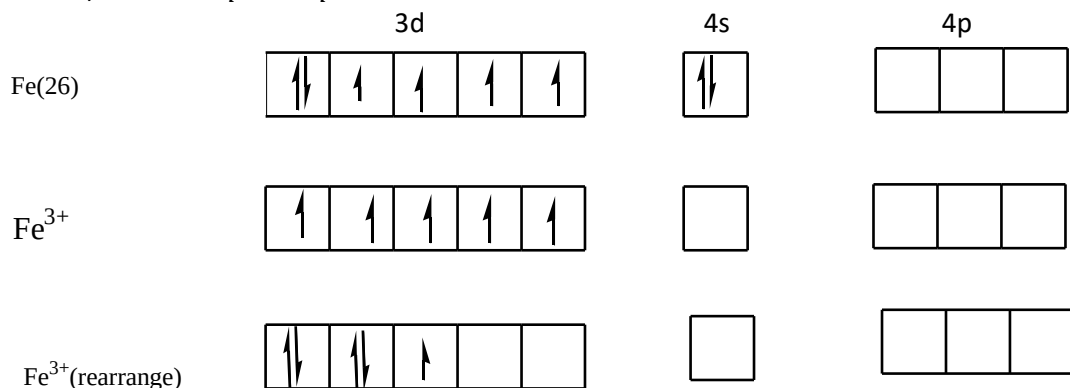
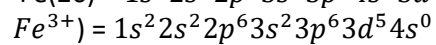
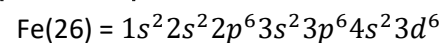
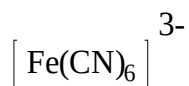
Answer:

The phenomenon of intermixing of atomic orbitals of slightly different energy to form new hybrid orbital is called hybridization. The two characteristics of hybridization are:

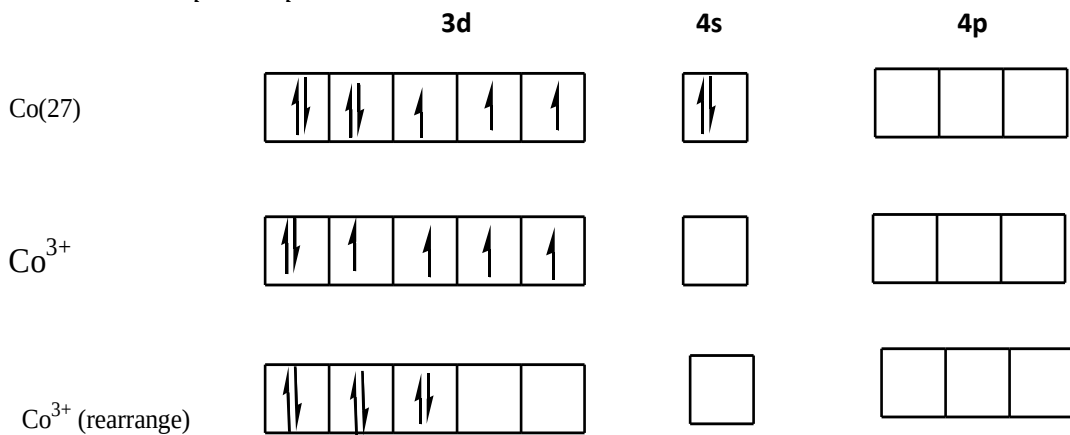
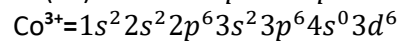
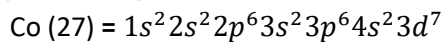
Those orbitals that take part in hybridization have almost equal or small energy difference and identical shape. Hybrid orbitals do not have pure property of s, p or d orbitals.

The shape of the important hybrid orbitals are given below:

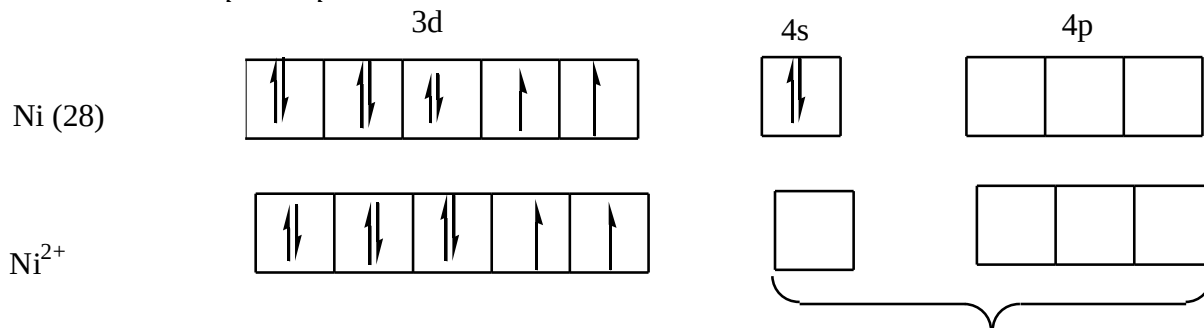
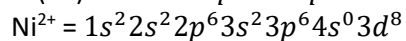
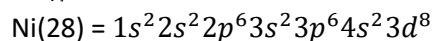
Co-ordination no.	Hybridization	Configuration
2	Sp	Linear
3	sp^2	Trigonal planar
4	sp^3 or sd^3	Tetrahedral
4	dsp^2	Square planar
5	dsp^3	Trigonal bipyramidal
6	d^2sp^3	Octahedral



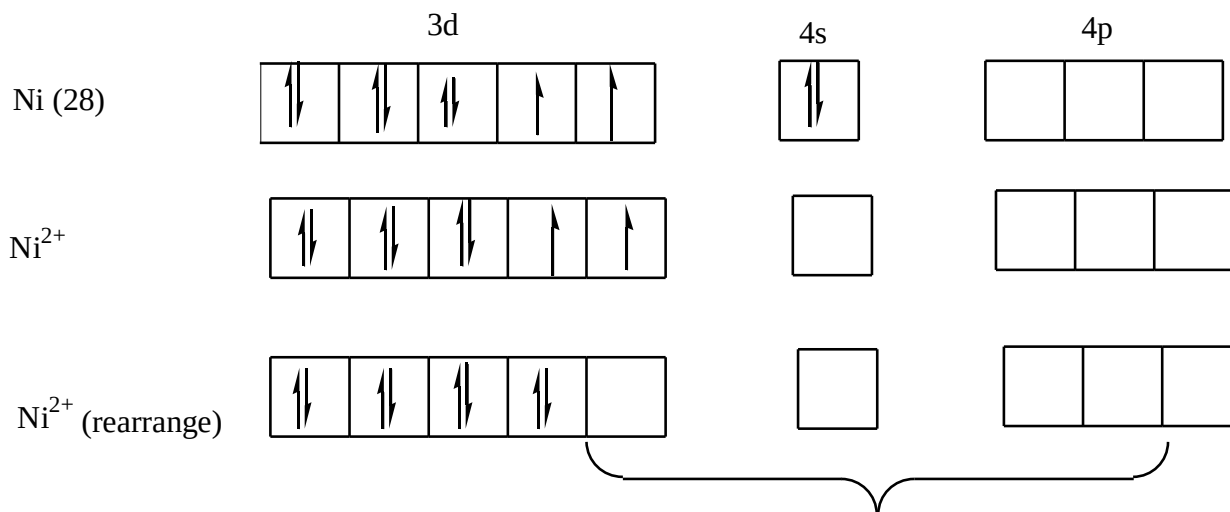
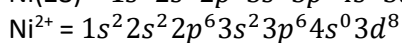
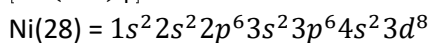
d^2sp^3 hybridization, Octahedral, 6 electron pairs from 6 CN^- ligand



d^2sp^3 hybridization, Octahedral, 6 electron pairs from 6 CN^- ligand



Sp^3 hybridization, Tetrahedral, 4 electron pairs from 4 Cl^- ligand



dsp^2 hybridization, square planar, 4 electron pairs from 4 CN^- ligand

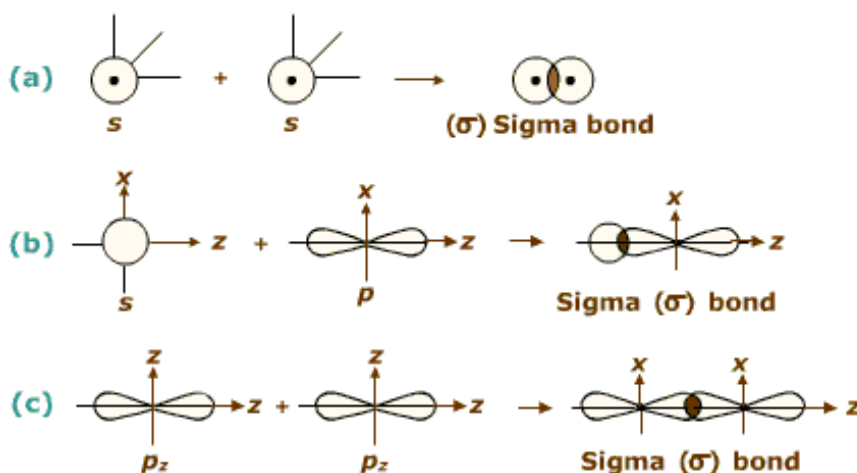
Question:

The complex $[\text{Ni}(\text{CN})_4]^{2-}$ is diamagnetic but $[\text{NiCl}_4]^{2-}$ is paramagnetic with two unpaired electron. Explain using VBT (Valence bond theory).

Diamagnetic materials are slightly repelled by magnetic fields. This is because all of the electrons are paired within the material. **Paramagnetic** materials are slightly drawn to the magnetic field.

Sigma bond:

A covalent bond which is formed between two atoms by the overlap of their half filled atomic orbitals along the line joining the nuclei of both the atoms is called a sigma bond. e.g.

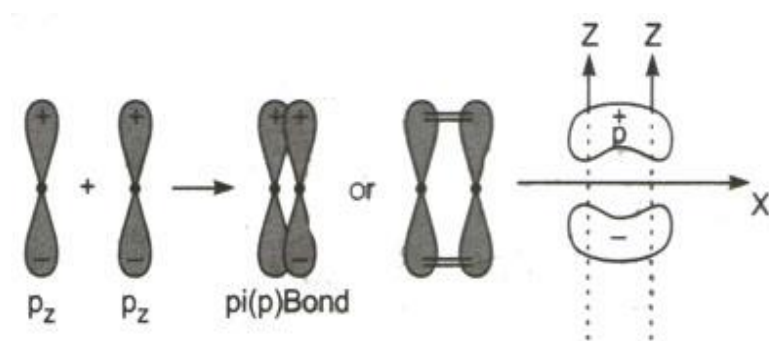


Formation of a sigma bond due to (a) The s - s overlap

(b) The s - p overlap (c) The p_z - p_z overlap

Pi bond:

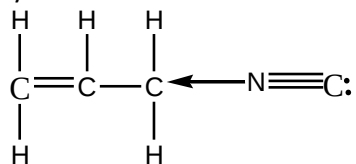
A covalent bond which is formed between two atoms by the overlap of their half filled p orbitals along a line perpendicular to their nuclear axis (side to side overlap) is called a pi bond. e.g.



Question: How many sigma(σ) and pi(π) bonds are there in allyl isocyanide?

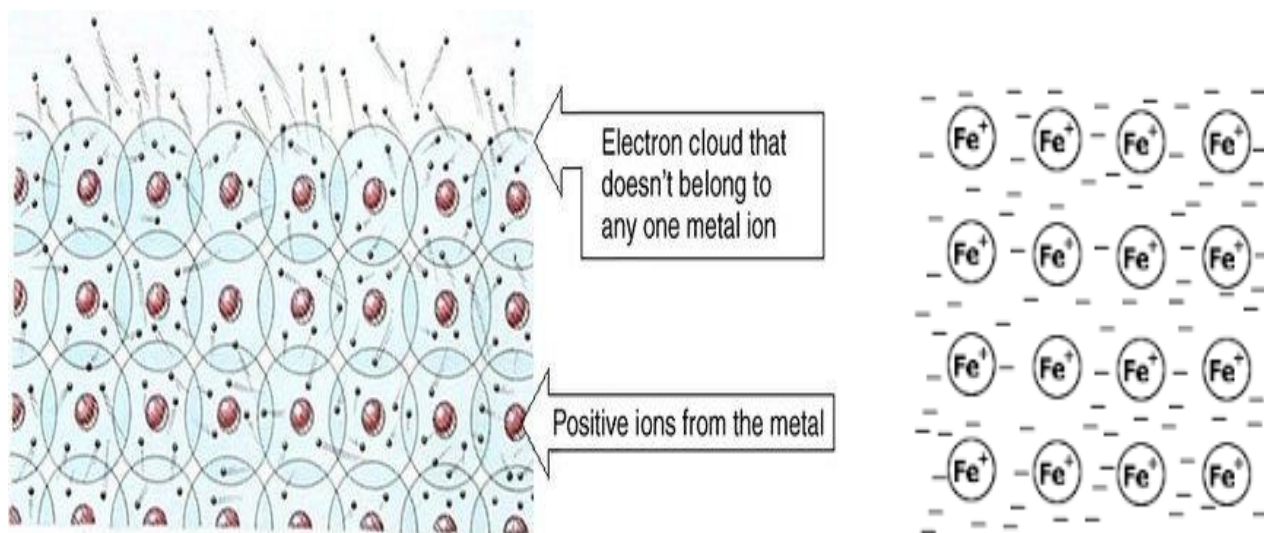
Answer:

The structure of allyl isocyanide molecule can be written as:

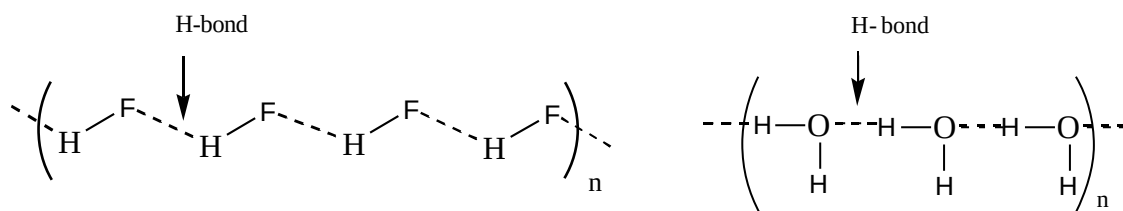


There are nine sigma and three pi bonds are present in allyl isocyanide.

Metallic bond: The peculiar type of bonding which holds the metal atoms together in a metal crystal is called metallic bond.

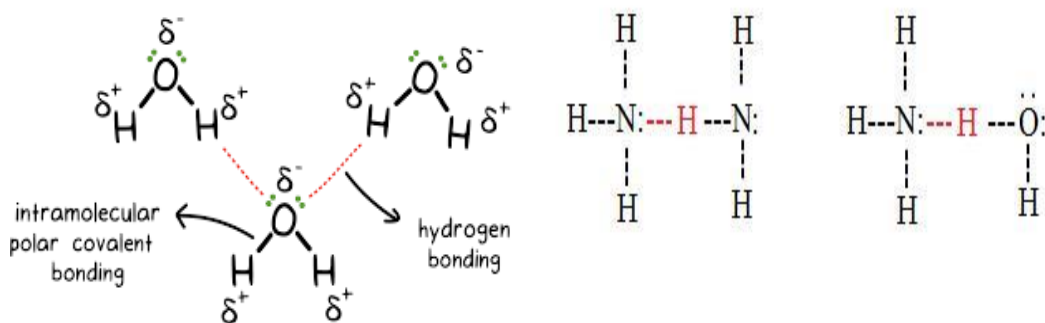


Hydrogen bond: The bond involves the bonding of a hydrogen atom with two strongly electronegative atoms (e.g. N, O, F) simultaneously.



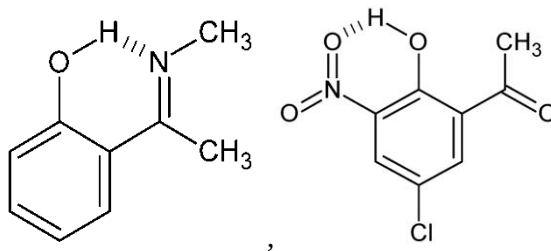
Intermolecular hydrogen bond

When the hydrogen bond is present between two atoms of two different molecules, then it is known intermolecular hydrogen bond.



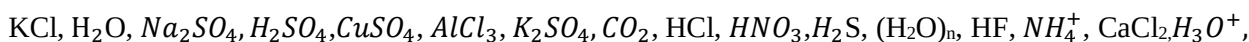
Intramolecular hydrogen bond:

When the hydrogen bond is present between two atoms of the same molecule, then it is known as intramolecular hydrogen bond.



Q-1. Indicate the different types of bond that are present in the following compounds:

Or, What types of bond are present in the following compounds?



Q-2. We know nitric acid is a strong acid which is harmful to human body. What types of bond are present in nitric acid?

Q-3. Name and represent the types of bonds present in $CuSO_4 \cdot 5H_2O$

Or, How many hydrogen bonded water molecules are there in $CuSO_4 \cdot 5H_2O$?

Q-4. For each of the following pairs of elements, State whether the binary compound formed is likely to be ionic or covalent, Give the formula and name of the compound.

a) Na, S b) Al, Cl c) Na, Cl d) H, Cl e) H, F

Q-5. Sodium amide, known commercially as sodamide, is used in preparing indigo, the dye used to color blue jeans. It is an ionic compound which the formula $NaNH_2$. What is the electron-dot formula of the amide anion, NH_2^- ?

Q-6. Phosphorous acid, H_3PO_3 , has the structure $(HO)_2PHO$, in which one H atom is bonded to the P atom and two H atoms are bonded to O atoms. For each bond to an H atom, decide whether it is polar or nonpolar.

Q-7. How many H-bonds are formed by each H_2O molecule in the structure of liquid water and how many water molecules are attached to each H_2O molecule and in what direction?

Q-8. Draw dots-and-crosses diagrams (showing outer electrons only) for the following covalent compounds.

a) ammonia, NH_3 b) hydrogen sulphide, H_2S c) hydrogen iodide, HI d) nitrogen trichloride, NCl_3
e) boron trifluoride, BF_3

Q-9. Carbon has the electronic structure $1s^2 2s^2 2p_x^1 2p_y^1 2p_z^0$. thinking about the bonding in methane:

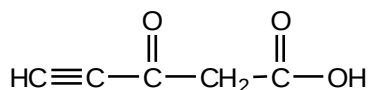
- The first step is promotion of an electron to give 4 unpaired electrons. Show this happening, in some combination of words and diagrams as you prefer.
- The second step is hybridization. Explain what this means, and what the result of it is.
- The final step is the formation of molecular orbitals involving carbon's orbitals and four hydrogen atoms.

Describe what happens at this stage.

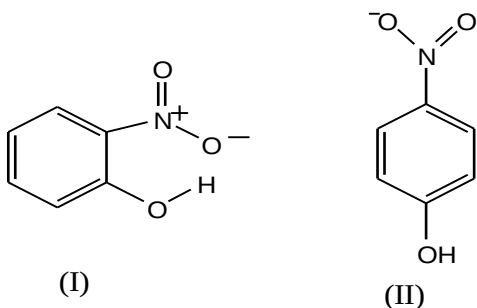
Q-10. By going through the same sort of process as you did in question 3, explain how the covalent bonding in the following compounds arises. In each case, you need only explain what happens to the central atom (the P, S or Xe).

a) PCl_5 b) SF_6 c) XeF_4

Q-11. Predict the hybridization of each carbon in the molecule of organic compound given below. Also indicate the total number of sigma and pi bonds of the molecule.

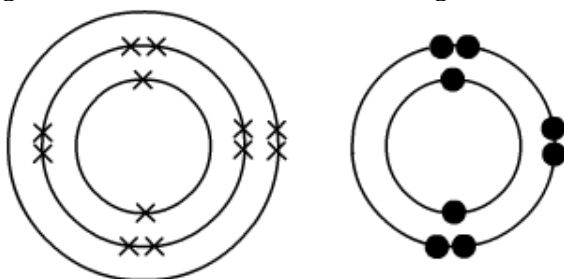


Q-12. Structures of molecules of two compounds are given below



Which of the two compounds will have the intermolecular hydrogen bonding and which compound is expected to show intramolecular hydrogen bonding?

Q-13. The diagram below shows the electron arrangements of magnesium and oxygen:



a) Draw a diagram showing how a bond is made between magnesium and oxygen.

b) What name is given to this type of chemical bond?

Q-14. a) What type of chemical bond would you expect in hydrogen fluoride, HF?

b) Draw a diagram to show how this bond is formed.

Q-15. Look at the diagram below:



a) Which type of structure is shown in the diagram?

b) Will a material with this structure be able to conduct electricity? Explain your answer.

Q-16. Look at the table below:

Substance:	Melting point (°C):
A	445
B	1907
C	-7

a) Which substance, A, B or C could have a metallic structure?

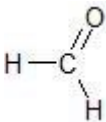
b) Which substance, A, B or C could have a simple covalent structure?

Q-17. What is an ionic bond? With two suitable examples explain the difference between an ionic and a covalent bond.

Q-18. For each pair, indicate which bond will be stronger.

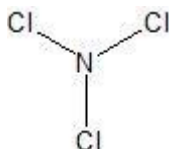
- a. C-H, Li-F b. Li-F, Mg-O c. Li-F, Cs-I

Q-19. Study the given Lewis structure and provide the following information:



- Draw in any lone pairs that are needed.
- How many bonding electrons does the carbon have? How many does Oxygen have?
- Is the octet of each atom in the molecule satisfied?
- How many single bonds are used in this molecule? How many double bonds?
- How many bonds are shared between the Carbon and Oxygen atoms? How many are shared between the Carbon and each Hydrogen?

Q-20. Answer each question regarding the following structure:



- Fill in all missing lone pairs
- How many bonding electrons does the Nitrogen have? How many non-bonding electrons does it have?
- How many bonding electrons do each Chlorine have? How many non-bonding electrons do they each have?
- How many electrons, in total, does Nitrogen share with all three Chlorines? Does each atom have an octet?
- Are the bonds formed in this molecule ionic or covalent?

Q-21. Why do elements on the same side of the periodic table tend to form covalent bonds? Why do elements on the opposite sides of the periodic table tend to form ionic bonds?

Q-22. A covalent bond consists of a pair of shared electrons between two atoms. How does this shared pair hold the two atoms together?

Q-23. Lithium aluminum hydride, LiAlH_4 , is an important reducing agent (an element or compound that generally has a strong tendency to give up electrons in its chemical reactions). Write the electron-dot formula of the AlH_4 ion.

Q-24. Give the Lewis formula for the arsenate ion AsO_4^{2-} . Write the formula of lead (II) arsenate.

Q-25. Give the Lewis formula for the selenite ion, SeO_3^{2-} . Write the formula of aluminum selenite.

Q-26. Iodic acid, HIO_3 , is a colorless, crystalline compound. What is the electron-dot formula of iodic acid?

Q-27. Selenic acid, H_2SeO_4 , is a crystalline substance and a strong acid. Write the electron-dot formula of selenic acid.

Q-28. Sodium, Na, reacts with element X to form an ionic compound with the formula Na_3X .

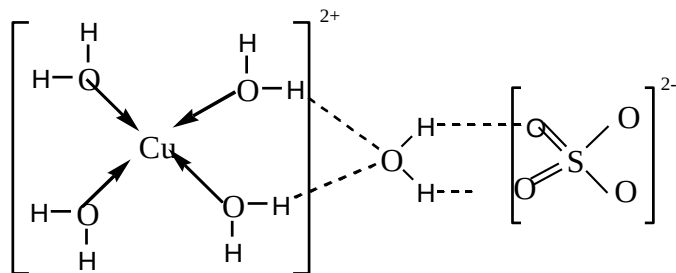
- What is the formula of the compound you expect to form when calcium, Ca, reacts with element X?
- Would you expect this compound to be ionic or molecular?

Answer

Q-3. Name and represent the types of bonds present in $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$

Or, How many hydrogen bonded water molecules are there in $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$?

Answer: The structure of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ is given below:



i) $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ has ionic bond between $[\text{Cu}(\text{H}_2\text{O})_4]^{2+}$ & SO_4^{2-} ions.

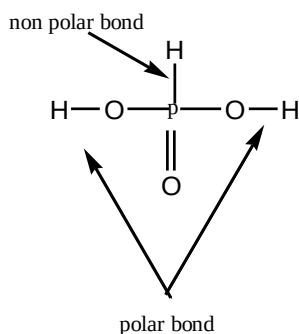
ii) This compound has 16 covalent bonds ($\text{O}-\text{H} = 10$, $\text{S}-\text{O} = 2$, $\text{S}=\text{O} = 4$)

iii) Four water molecules are linked with Cu^{2+} ion in coordination sphere by 4 co-ordinates covalent bonds. ($\text{H}_2\text{O} \rightarrow \text{Cu}$)

iv) The remaining H_2O molecule forms four H-bonds. Two H-bonds are formed by two H-atoms with two O- atoms of SO_4^{2-} ion and two H-bonds are formed by O-atom with H-atoms of two co-ordinated H_2O molecules present in co-ordination sphere

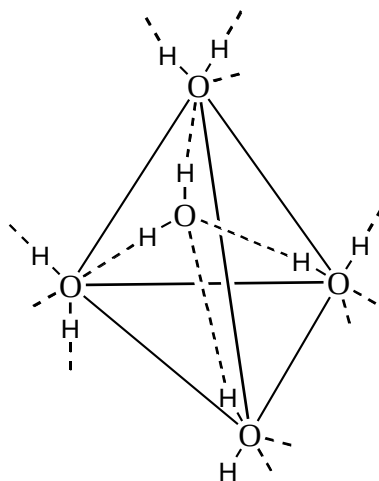
Q-6. Phosphorous acid, H_3PO_3 , has the structure $(\text{HO})_2\text{PHO}$, in which one H atom is bonded to the P atom and two H atoms are bonded to O atoms. For each bond to an H atom, decide whether it is polar or nonpolar.

Answer: The structure of H_3PO_3 is given below:



Q-7. How many H-bonds are formed by each H_2O molecule in the structure of liquid water and how many water molecules are attached to each H_2O molecule and in what direction?

Answer: The structure of liquid water is given below



Each H_2O molecule forms four H-bonds: two H-bonds are formed by O atom of H_2O molecule and two H-bonds are formed by two H-atoms of the same H_2O molecule.

Each H_2O molecule is linked tetrahedrally with four H_2O molecules through H-bonds.

Q-15.. Look at the diagram below:



a) Which type of structure is shown in the diagram?

b) Will a material with this structure be able to conduct electricity? Explain your answer.

Answer:

a) In the diagram the structure is metallic. **b)** The material will be able to conduct electricity because there are free electrons that are able to move through the structure.

Q-16. Look at the table below:

Substance:	Melting point ($^{\circ}C$):
A	445
B	1907
C	-7

a) Which substance, A,B or C could have a metallic structure?

b) Which substance, A, B or C could have a simple covalent structure?

Answer: **a)** B (highest melting point) **b)** C (lowest melting point)

Q-18. For each pair, indicate which bond will be stronger.

a. C-H, Li-F

b. Li-F, Mg-O

c. Li-F, Cs-I

Answer:

a. Li-F is stronger because it is an ionic bond while C-H is a covalent bond.

b. Mg-O is stronger because Mg is charged $2+$ and O is charged $2-$ while each ion in Li-F has only a charge of magnitude 1. Ions with higher charges are more strongly bound.

c. Li-F is stronger than Cs-I because the Li-F bond is far shorter than the Cs-I bond. By Coulomb's law, shorter ionic bonds should be stronger.